

Project 2: Back-End Design

Requirements:

Practice the topic of back-end design for a MIPS16 processor:

(Same teams as project 1).

Using **ICC** on the provided VM and starting from the provided **MIPS16 design** at /home/ahesham/ref_flow_before, apply **all steps** you learned of the back-end design all the way to generate a **complete chip layout**. Take the **provided reference script** as your starting point to go through and optimize each step of the flow and reach the final layout that meets the following specs.

Area smaller or equal to 100000 square microns. **Clean** DRC report.

Frequency greater or equal to 400 MHz. **Clean** setup/hold timing report.

The smaller the area and the higher the frequency the higher your mark will be, i.e. Target minimum area-delay product (area/frequency).

Deliverables:

A single zip file **<Gxx_project2>.zip** containing a top-level directory named <Gxx> under which **only TWO** components:

- I. A single pdf file named <Gxx_project2>.pdf containing the following:
 1. Cover page. Contains: project title + students name + emails + department
 2. Chapter 1: Introduction. Contains project idea.
 3. Chapter 2: Steps of back-end design. Contains sections for each step of the flow, where each section provides as much description of your effort as possible, along with the final reports and/or snapshots highlighting key info, as well as, your conclusions.
 4. Chapter 3: Conclusions. Contains final comments on your progress meeting the required specs.
- II. A directory named <files> containing your script and all report files.
xx is your 2-digit team group number as identified by the TA.

Submission:

Submit to course LMS site before end of day 27th of December 2025.

All members of the group must submit the same file on LMS.